

Habib University



Course Title: Digital Logic and Design (EE-172 / CS-130 / CE-222)

Instructors: Farhan Khan, Saad Baig, Ahmed Ali

Assignment No. 01

Release Date: Jan 24, 2025

Due by: Jan 31, 2025 11:59 PM

Total points: 100

Points obtained:

Student Name:	Student ID:	Section:
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Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; representing complements of a number; and performing BCD arithmetic.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in a reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply r-base (binary, octal, decimal, hexadecimal numbers systems) to digital systems and carry out arithmetic operations and conversions	✓
CLO 2	Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)	
CLO 3	Design combinational logic circuits using logic gates	
CLO 4	Design sequential systems using finite state machine methodology	

P#	Questions	Pts																																						
1	Convert the following numbers with the indicated bases to decimal: a) $(71.67)_8$ b) $(122)_3$ c) $(65A)_{11}$ d) $(0.435)_6$	2x4																																						
2	Convert the following numbers a) $(50.08)_{10}$ to octal b) $(0.6775)_8$ to hexadecimal c) $(715)_{10}$ to base-13 d) $(110011.011)_2$ to decimal	2x4																																						
3	A 12-bit register stores 100010010111. Determine what value is stored in decimal in the register if the bits represent: a) Unsigned number b) Signed number (2's complement system) c) Signed number (1's complement system) d) BCD	2x4																																						
4	You are an artist in a secret society known for creating encrypted messages within their artwork. The society has decided to use a unique symbol system to encode the date when their next exhibition will take place. The exhibition date is a closely guarded secret, and only society members will be able to decode it. You have been tasked with creating a system to represent the date using the following characters: (You are required to use all the following characters only (no other characters are allowed)): <table border="1"><tr><td>@</td><td>*</td><td>\$</td><td>%</td><td>^</td><td>/</td><td><</td><td>></td></tr></table> a) Create a suitable number system and fill in the following table up to 12. State your assumptions clearly. <table border="1"><tr><td>Actual</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>...</td></tr><tr><td>Coded</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>...</td></tr></table> b) Represent the year 1601_{10} in your coded notation c) In your number system, what does “(\$@*)” represent in decimal?	@	*	\$	%	^	/	<	>	Actual	0	1	2	3	4	5	6	7	8	9	10	11	12	...	Coded														...	4+2+2
@	*	\$	%	^	/	<	>																																	
Actual	0	1	2	3	4	5	6	7	8	9	10	11	12	...																										
Coded														...																										
5	Suppose you are working in a tech company that develops a software tool, and the tool requires converting a number for its processing. The problem is that the conversion process is very slow, and your task is to optimize it:	4																																						

	The decimal number 975 had to be converted into binary to feed into the computer and you have the following two ways: • Convert directly to binary • Convert first to hexadecimal and then from hexadecimal to binary. Which method is faster? Justify your answer with reasoning. (Requirement: You must show the working of both the parts and then give your reasoning based on time and method used. Note: Marks will only be awarded if you can justify your answer).																																																					
6	In an ancient kingdom, the royal treasury used a base 20 (vigesimal) system for all financial transactions. The kingdom's bankers, who were experts in mathematics, issued loans and collected repayments based on this unique system. a) Extend the digit representation used in the hexadecimal number system to form the base 20 digits. b) Person A, the kingdom's chief banker, lends person B a sum of $(DHJ5F)_{20}$ to fund a royal expedition. If B repays the loan within 6 lunar months, the interest is calculated as $(21DHH)_{20}$. If the expedition is delayed and B does not repay the loan on time, an additional interest of $(5B3E)_{20}$ is added for each lunar month B fails to return the loan. If B returns the loan 8 lunar months after borrowing it, calculate the total amount B must repay to Person A c) What is the net gain or loss for Person A, expressed in the decimal (base 10) system & Binary (Base 2) system.	2+5+3																																																				
7	Convert the last three alphabets of your first name and the first three alphabets of your last name to hexadecimal system, according to the given table. <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td><td>L</td><td>M</td></tr><tr><td>21</td><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td><td>29</td><td>2A</td><td>2B</td><td>2C</td><td>2D</td></tr></table> <table><tr><td>N</td><td>O</td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td><td>U</td><td>V</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>2E</td><td>2F</td><td>30</td><td>31</td><td>32</td><td>33</td><td>34</td><td>35</td><td>36</td><td>37</td><td>38</td><td>39</td><td>3A</td></tr></table> For example, your name is “ALI HUSSAIN,” you will convert “ALI” from ALI and “HUS” from HUSSAIN as follows. (Use uppercase letters) “ALI” ➔ 212C29 “HUS” ➔ 283533 (Let the first number be X1 and the last name be X2) a) Using 16’s Complement method, perform the following: $X1 - X2$ b) Using 15’s Complement method, perform the following: $X2 - X1$ c) Convert both X1 and X2 to radix 8 and find their sum	A	B	C	D	E	F	G	H	I	J	K	L	M	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	2E	2F	30	31	32	33	34	35	36	37	38	39	3A	2x2+4
A	B	C	D	E	F	G	H	I	J	K	L	M																																										
21	22	23	24	25	26	27	28	29	2A	2B	2C	2D																																										
N	O	P	Q	R	S	T	U	V	W	X	Y	Z																																										
2E	2F	30	31	32	33	34	35	36	37	38	39	3A																																										
8	Using the r’s complement method, where r is the radix of the numbers mentioned, find the following differences:	3x2																																																				

	a) $(11043)_{16} - (3289)_{16}$ b) $(11001.101)_2 - (10111.001)_2$																													
9	Convert the following decimal numbers to BCD to solve: a) $0.493 + 1.965$ b) $79.84 - 34.51$ c) $1357 + 870$ d) $0.721 + 1.48$	2x4																												
10	Convert the following to signed (2's Complement) 6-bit binary numbers and perform the given operations. Also mention if there is an overflow or not. a) $(+12)_{10} + (-1)_{10}$ b) $(-7)_{10} - (-8)_{10}$ c) $(+10)_{10} - (+30)_{10}$ d) $(-15)_{10} - (-9)_{10}$	2x4																												
11	What is the smallest binary integer that can be expressed with a 32-bit computer using 2's complement signed representation? What are the equivalent decimal and hexadecimal numbers?	2x3																												
12	Complete the following table using 8-bit binary numbers: <table><tr><th>Decimal</th><th>Signed Magnitude</th><th>Signed 1's Complement</th><th>Signed 2's Complement</th></tr><tr><td>-108</td><td></td><td></td><td></td></tr><tr><td>-128</td><td></td><td></td><td></td></tr><tr><td>-23</td><td></td><td></td><td></td></tr><tr><td>+50</td><td></td><td></td><td></td></tr><tr><td>+122</td><td></td><td></td><td></td></tr><tr><td>+0</td><td></td><td></td><td></td></tr></table>	Decimal	Signed Magnitude	Signed 1's Complement	Signed 2's Complement	-108				-128				-23				+50				+122				+0				1x6
Decimal	Signed Magnitude	Signed 1's Complement	Signed 2's Complement																											
-108																														
-128																														
-23																														
+50																														
+122																														
+0																														
13	Find out the even parity bit of the following binary numbers: I. 00010110 II. 10100101	1x2																												
14	Convert the following: a) BCD number 1001010100110111 to excess-3 b) Binary 01110110001011000 to Gray Code	10																												